

Biodegradable PVA-Gelatin Films Coated with Reduced Graphene Oxide: Effects on Electrical Conductivity and Mechanical Durability

RSEF Category: Materials Science (MATS)

Abstract

In biomedical research, damaged tissue and organs are repaired through tissue scaffolding and organ regeneration. However, these techniques demand the use of compatible biomaterials with strong mechanical properties, such as elasticity, durability, and tensile strength, and strong electrical properties like conductivity. Utilizing PVA-gelatin polymers, this study aims to focus on whether incorporating increased concentrations of reduced graphene oxide (rGO), 0.0 wt%, 0.1 wt%, 0.5 wt%, and 1.0 wt%, can improve mechanical durability and electrical conductivity, making it an optimal component for biomaterials in biomedical applications. Conductivity tests and load limit tests were used to assess conductivity and durability/tensile strength, respectively. Through comparing tensile strength ratios and performing ANOVA statistical analysis on the conductivity values across the different concentrations of rGO in the PVA-g polymers, it was found that while higher rGO concentrations displayed better conductivity and strength than lower ones, no rGO concentrations could surpass the negative control of zero rGO wt%. These findings best support that rather than improving the electrical and mechanical properties of biodegradable polymers such as PVA-g, rGO inhibits these qualities, making it less practical for biomedical applications. In future work, exploring alternative additions such as carbon nanotubes (CNTs) and surface-localized nanocomposites may prove more potential for enhanced electrical conductivity and mechanical durability. With stronger alternatives such as these, the challenges of tissue engineering and organ generation can be met with new and improved biomaterials that address the electrical and mechanical demands of the biomedical field in the modern era.

Introduction

In biomedical research, tissue scaffolding and organ regeneration are powerful techniques that can address biomedical challenges such as replacing and/or repairing damaged tissue and organs.

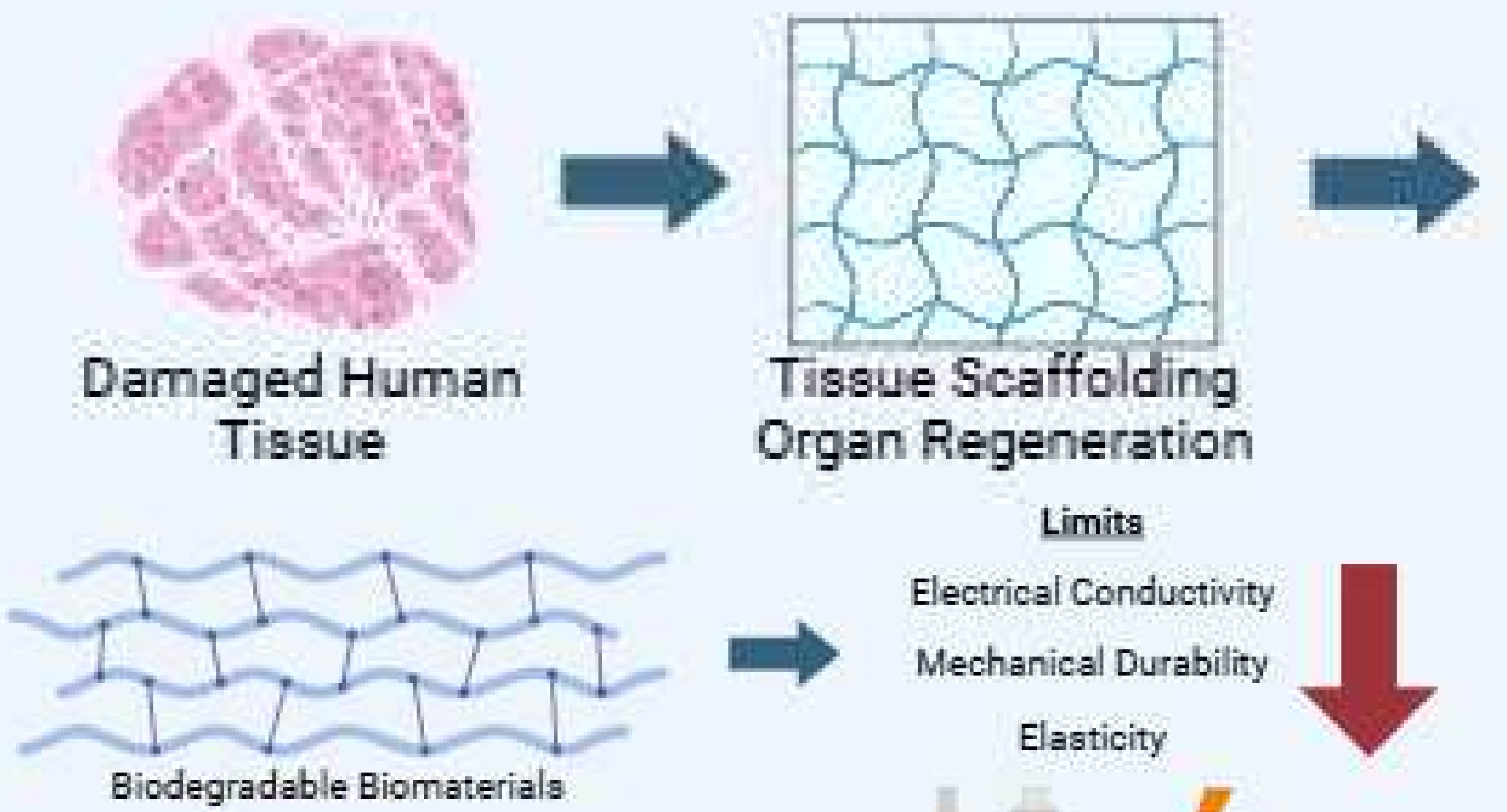


Figure 1: Background Visualization. Created in BioRender: <https://biorender.com>

However, while these techniques are powerful and serve as appropriate responses to the challenges of modern biomedical research, they are often limited due to requirement for certain materials. Specifically, biodegradable biomaterials with high levels of mechanical durability, elasticity, and electrical conductivity are needed to ensure methods such as scaffolding, organ regeneration, and other tissue engineering methods are effective. Since many biodegradable biomaterials prove to have minimal electrical and mechanical properties, this is often a challenge. One that is best addressed by implementing the addition of certain compounds that address the electrical and mechanical gap within biodegradable biomaterials or polymers.

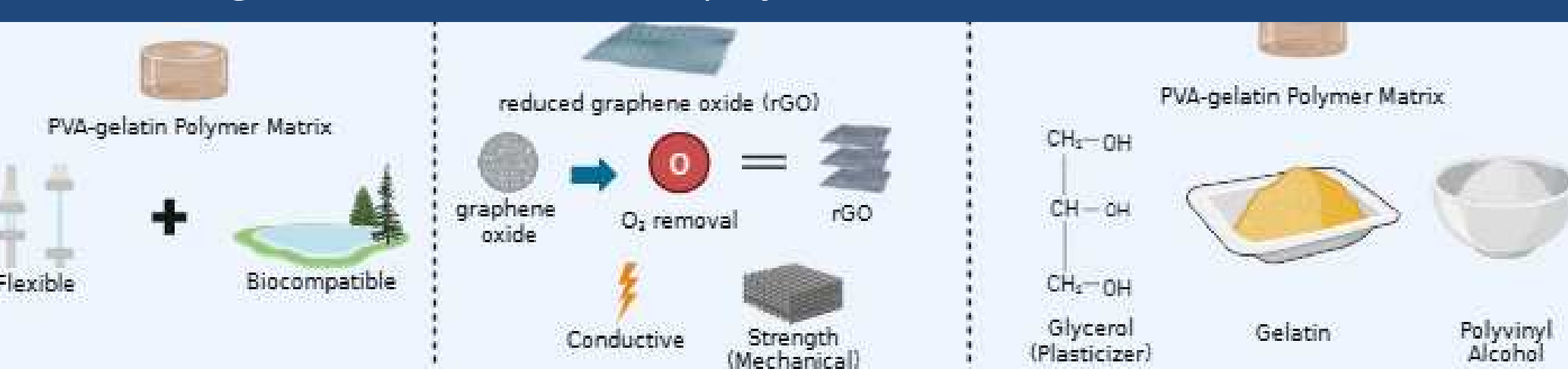


Figure 2: Materials Visualization. Created in BioRender: <https://biorender.com>

Using a polymer base of polyvinyl alcohol (PVA) powder and gelatin, this study aims to observe whether incorporating higher concentrations of reduced graphene oxide (rGO) can improve the mechanical durability and electrical conductivity of biomaterials. It is hypothesized that this is possible because of rGO's ability to improve mechanical properties in PVA-based compounds and its electrical benefits in gelatin-based ones (Wang et. al., 2012; Yin et. al., 2022).

Methodology

Independent Variable

Concentration of rGO (reduced graphene oxide) added to PVA-gelatin films

Dependent Variable

Electrical conductivity in voltage and stress-strain ratios of mechanical durability

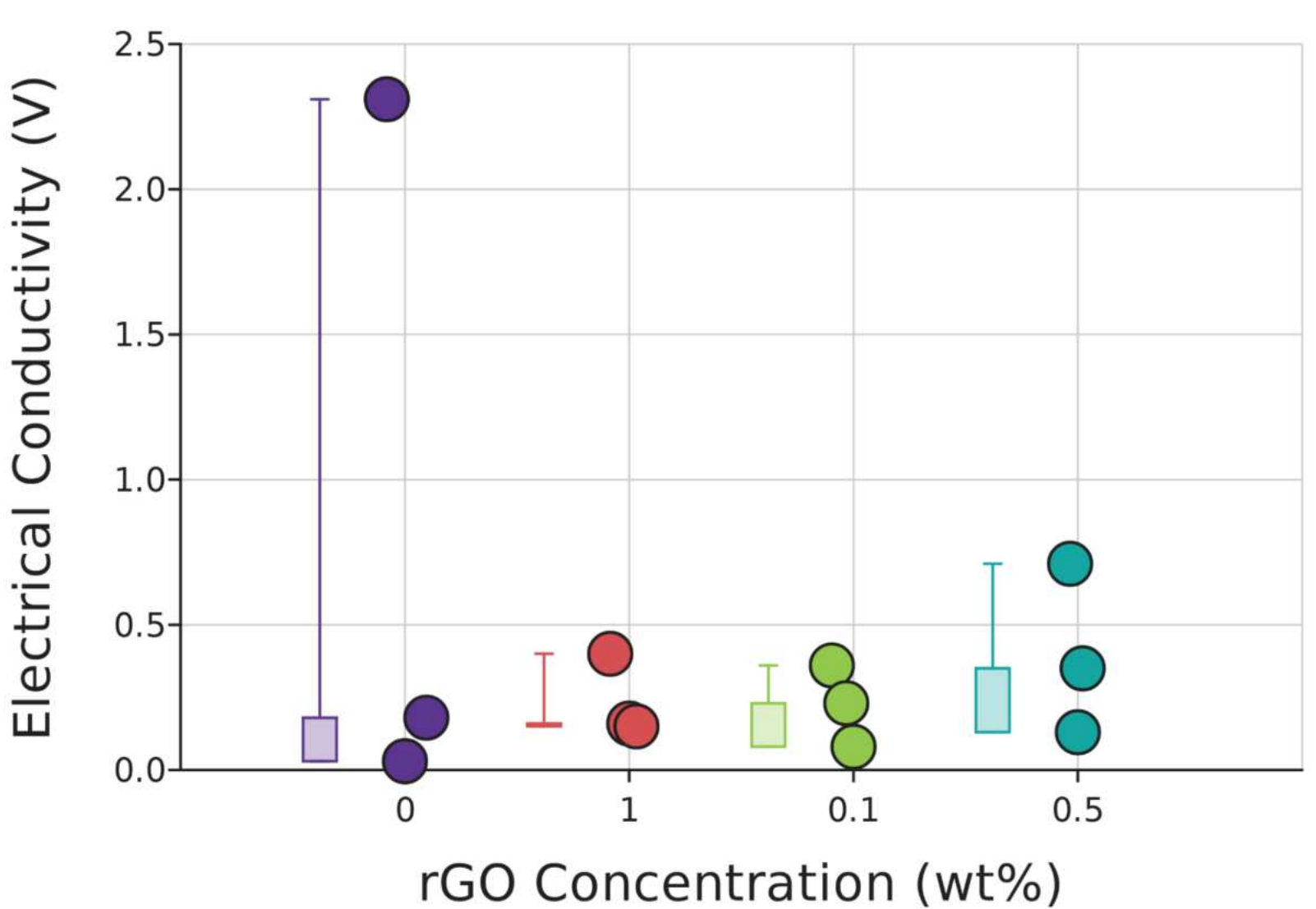
rGO Weighting (wt%)

0.0, 0.1, 0.5, and 1.0

Assays

- Electrical Conductivity Testing
- Load Limit Testing (via Mark 10 F305 Force Stand)

Data & Results



Note: Box and whiskers plot shows the median value (line), interquartile range (box), and the extent of the data (whiskers above and below at min / max data points).

Figure 4: Median-based graph comparing electrical conductivity across different concentrations of rGO in PVA-gelatin films.

Electrical Conductivity Data			
Graphene Oxide Concentration (wt%)	Electrical Conductivity (V)	Resistivity (Ω-m)	Sheet Resistance (Ω per square)
0	2.52	39.321	4915.224
0.1	0.67	19.626	3270.934
0.5	1.19	11.958	1992.984
1	0.71	15.417	2202.381

Table 2: Chart outlining the average conductivity values across three trials of electrical conductivity testing for each level of rGO concentrations embedded in the PVA-gelatin polymers.

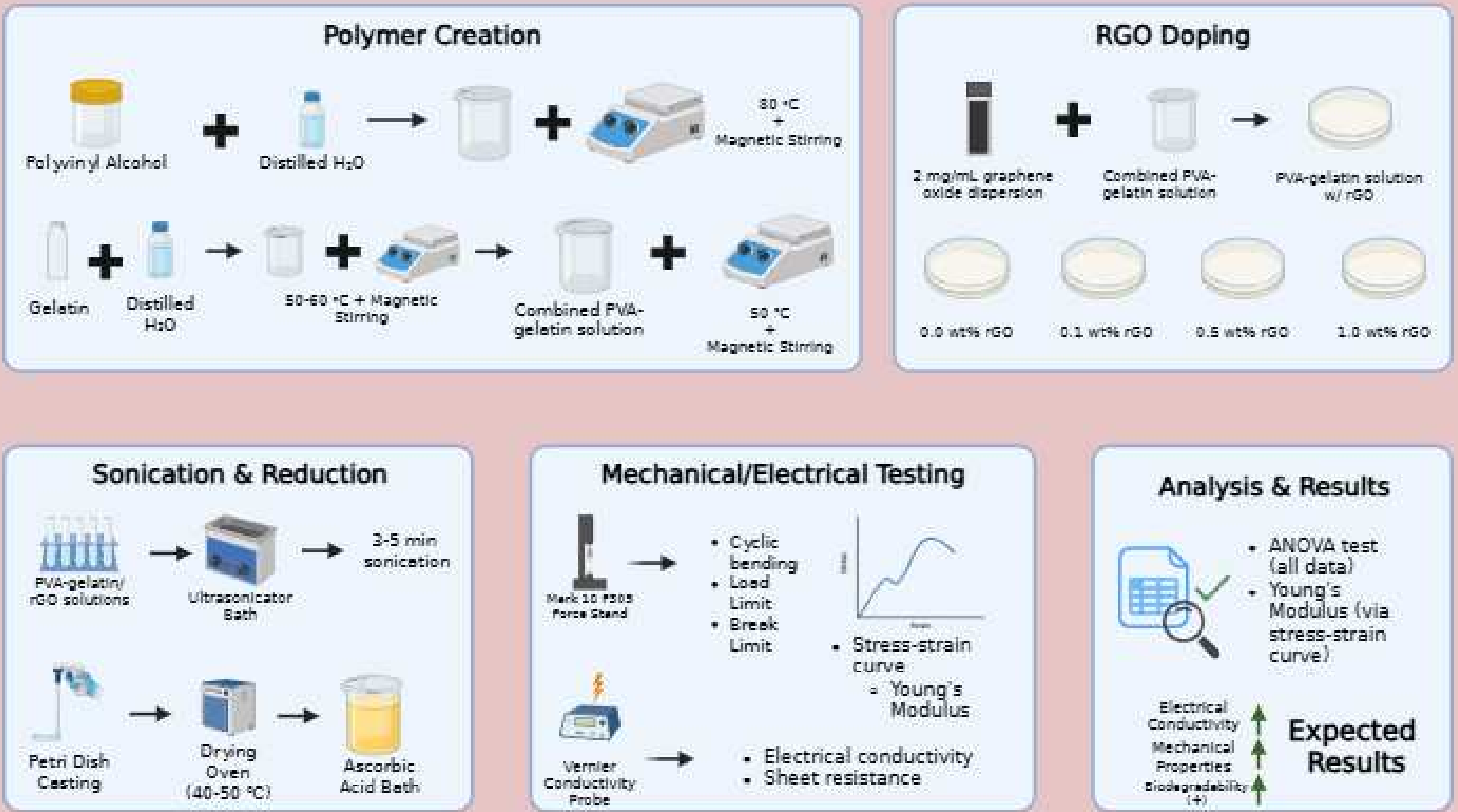


Figure 3: Methodological Visualization. Created in BioRender: <https://biorender.com>

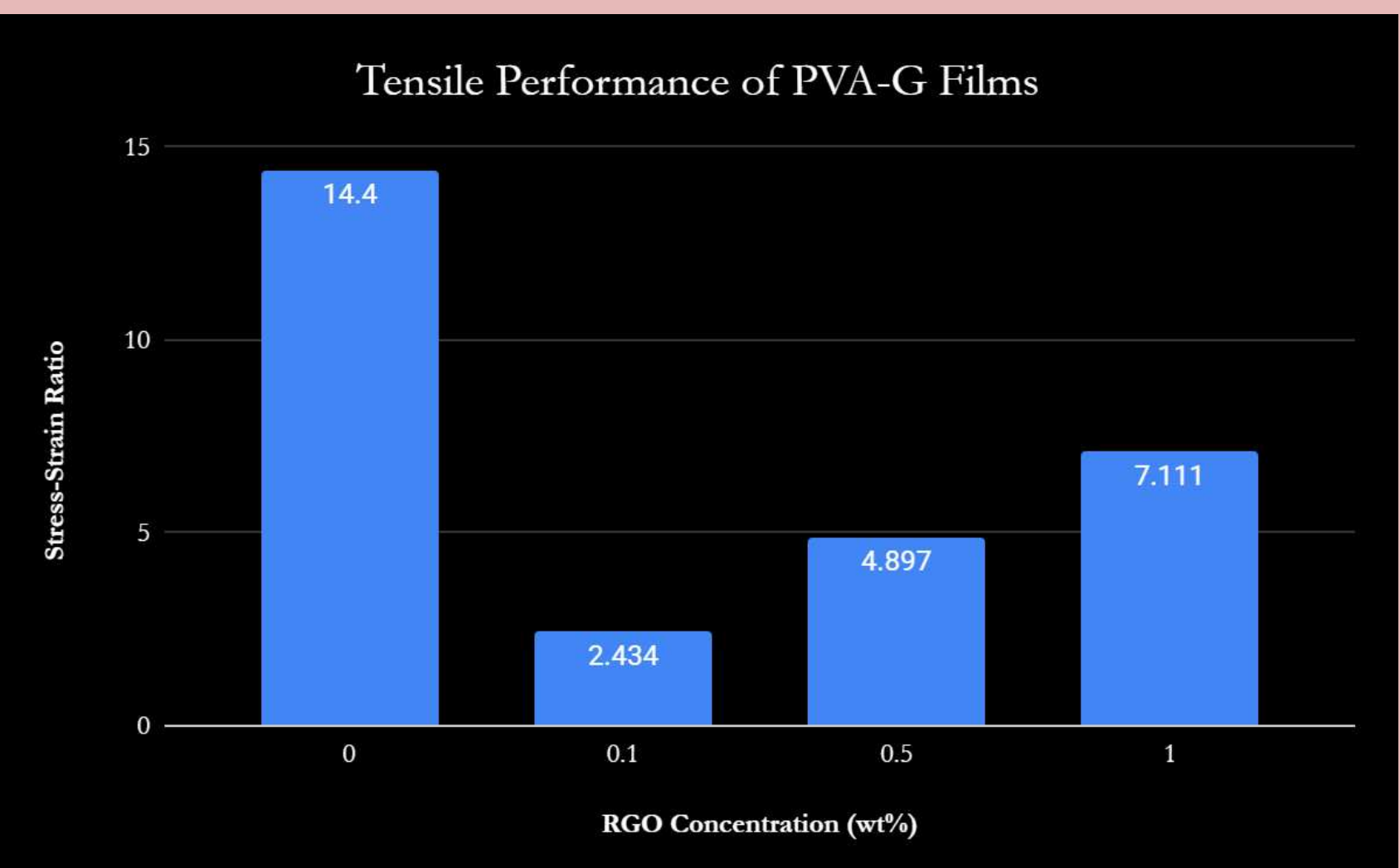


Figure 5: Bar graph outlining the tensile performance of the polyvinyl alcohol-gelatin (PVA-G) films exposed to different concentrations of reduced graphene oxide (rGO). Tensile performance is based on the calculated stress-strain ratio: the higher the ratio, the better the overall performance.

Mechanical Assay: Trial 1 Overview	
[rGO] (wt%)	Young's Modulus (σ:ε)
0	14.4
0.1	2.434
0.5	4.897
1	7.111

Table 1: Chart outlining the different Young's Modulus (stress-strain ratios) across the different PVA-gelatin films with their respectively different rGO concentrations.

Effect	Degrees of Freedom (df)	F-statistic (MS / MS residual)	P-value	Interpretation of P
GO Concentration (wt%)	3	0.57	0.65	A P-value of 0.65 means no evidence that the groups might be different.
Error or Residual	8			

Figure 6: Results of the ANOVA statistical analysis test used to analyze the electrical testing data across all three trials.

Discussion

The p-values, which depending on their value from 0.01 to 1.00, identify the statistical difference between the groups as a higher p-value reflects lower statistical difference. As can be seen in Table 2, the electrical conductivity values appear to readily decrease as rGO concentrations in the PVA-gelatin polymers increase. Furthermore, as depicted in Figure 6, the p-value of 0.65 reflects that there was statistically little-to-no difference between the concentration of rGO and the median electrical conductivity of the films. In regards to the electrical properties of the PVA-gelatin polymers, it is best understood that rGO was not able to improve electrical conductivity at higher concentrations, as the control group of 0.0 wt% rGO (no rGO presence) displayed higher conductivity than any concentration of rGO. In regards to mechanical durability, the Young's Modulus values of each PVA-gelatin polymers across one trial of mechanical testing were compared. Young's Modulus (the direct stress-strain ratio of the polymers) was found by graphing stress-strain curves and taking the slope of the initial linear portion of the graph. Comparison of these different YM values, it was found that rGO was incapable of improving mechanical durability as the stress-strain ratio of the control group was higher than that of the rGO concentrations (despite each rGO concentration showcasing a higher YM value than the previous one with the exception of the control group). To improve electrical and mechanical properties, alternatives to rGO should be explored.

Future Work

As an alternative to the use of reduced graphene oxide (rGO), carbon nanotubes (CNTs) and surface-localized nanocomposites (SLNCs) of other viable biomaterials. While rGO failed to showcase statistically significant improvement in electrical conductivity and mechanical durability, incorporating CNTs and SLNCs may prove more potential (Ryan et al., 2023).

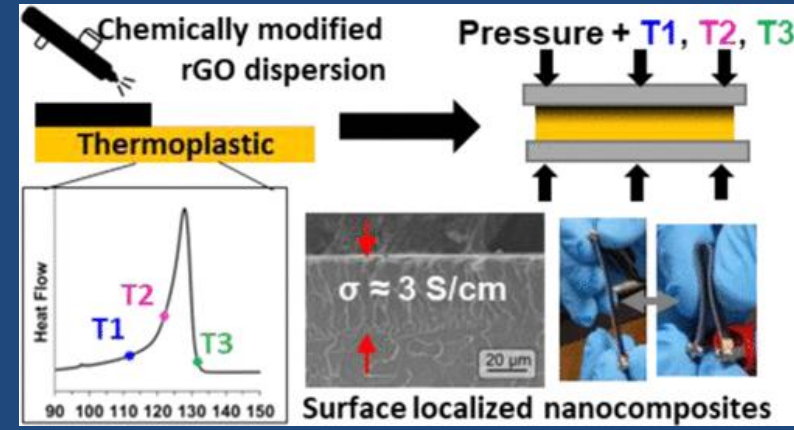


Figure 7: Diagram outlining the incorporation of surface-localized chemically modified rGO applied for electrical conductivity benefits.

References

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